Int. J. Radiation Oncology Biol. Phys., Vol. 51, No. 1, pp. 138–147, 2001

0360-3016/01/$–see front matter

ELSEVIER

PII S0360-3016(01)01560-7

CLINICAL INVESTIGATION **Eye**

MAXIMIZING LOCAL TUMOR CONTROL AND SURVIVAL AFTER PROTON BEAM RADIOTHERAPY OF UVEAL MELANOMA

EMMANUEL EGGER, PH.D.,* ANN SCHALENBOURG, M.D.,*† LEONIDAS ZOGRAFOS, M.D.,† LUDMILA BERCHER, M.D.,† TERENCE BOEHRINGER, PH.D.,* LINE CHAMOT, M.D.,† AND GUDRUN GOITEIN, M.D.*

*Division of Radiation Medicine, Paul Scherrer Institute, Villigen, Switzerland; †Hôpital ophtalmique Jules Gonin, Lausanne, Switzerland

Purpose: This study reports local tumor control and survival after proton beam radiotherapy (PBRT) of uveal melanoma. It identifies the risk factors for local tumor-control failure and for ocular tumor-related death. It presents the improvements implemented to increase the rate of local tumor control, and compares the survival rate of patients with locally controlled tumors to those of patients who had to receive a second treatment.
Patients and Methods: We have treated 2,435 uveal melanomas with PBRT between March 1984 and December 1998. Data were analyzed as of September 1999. Patients' age ranged from 9 to 89 years; there were 1,188 men and 1,247 women. The largest tumor diameter ranged from 4 to 26 mm, and tumor thickness from 0.9 to 15.6 mm. Median follow-up time was 40 months.
Results: Local tumor control probability at 5 years was improved from 90.6 ± 1.7% for patients treated before 1988, to 96.3 ± 0.6% for patients treated between 1989 and 1993, and became 98.9 ± 0.6% for patients treated after 1993. Among 2,435 treated patients, 73 (3%) had to receive a second treatment because of tumor regrowth. Cause-specific survival at 10 years was calculated to 72.6 ± 1.9% for patients with controlled tumors compared to 47.5 ± 6.5% for those with recurrent tumors.
Conclusion: Reduced safety margins, large ciliary body tumors, eyelids within the treatment field, inadequate positioning of tantalum clips, and male gender were identified to be the main factors impairing local tumor control. The improvement of local tumor control rate after 1993 is attributed to changes implemented in the treatment procedure. Our data strongly support that the rate of death by metastases is influenced by local tumor control failure: improvement of the local tumor control rate results in a better survival rate.

Uveal melanoma, Proton beam radiotherapy, Results, Local tumor control, Survival.

INTRODUCTION

Uveal melanoma is the most common primary intraocular malignant tumor. Ophthalmologists are familiar with this disease entity, and several therapeutic options are available today in specialized centers. These include enucleation, local resection, brachytherapy using various radioactive isotopes (e.g., cobalt-60 [Co-60], iodine-125 [I-125], and ruthenium-106 [Ru-106]), external beam radiotherapy with photons from linear accelerators or gamma knives, and charged particle beams such as helium ions and protons (1–20). The low incidence of uveal melanoma and the diversity of its treatment options make it difficult to gather data on long-term results for a large number of equally treated patients. Most reports do not have more than 1–2 years of follow-up and do not include more than 300 patients. Long-term results are important because uveal melanomas are slow-growing tumors. Metastases have been reported to appear after more than 20 years following enucleation (21). At our center, we have treated a recurring uveal melanoma that had received brachytherapy with a Co-60 applicator 18 years earlier.

Our study is based on 15 years of experience in the treatment of 2,435 uveal melanomas with proton beam radiotherapy (PBRT). We report on the number of patients with documented tumor regrowth following PBRT, and identify the probable risk factors for these recurrences. We also describe the changes to the treatment procedure to improve the rate of local tumor control. The correlation between local tumor control and survival is analyzed. Finally, our report investigates the tumor- and patient-related factors that influence death from metastases.

PATIENTS AND METHODS

This study reports on 2,435 cases of uveal melanoma in 2,432 patients treated with PBRT at the Paul Scherrer In-

Reprint requests to: Emmanuel Egger, Division of Radiation Medicine, Paul Scherrer Institute, CH - 5232 Villigen – PSI, Switzerland. Tel: ++41 56 310 40 94; Fax: ++41 56 310 35 15; E-mail: emmanuel.egger@psi.ch
Accepted for publication 5 March 2001.

Table 1. Country of residence of treated patients

	No. Patients		Lost to follow-up		No. of follow-up examinations	
Country	No.	%	No.	%	Total	% done in Lausanne
Switzerland	431	17.72	4	0.9	2,304	90.2
Italy	1,123	46.17	29	2.5	5,468	92.9
France	260	10.69	22	8.5	1,665	69.8
Greece	224	9.21	98	43.7	540	65.0
Spain	95	3.91	22	23.2	394	80.5
Belgium	56	2.30	1	1.8	242	54.1
Germany	52	2.14	4	7.7	250	70.0
Austria	23	0.95	1	4.3	93	59.1
Netherlands	37	1.52	2	5.4	112	59.8
Portugal	55	2.26	6	10.9	160	89.4
Other countries	76	3.13	15	19.7	193	62.2
Total	2,432	100	204	8.4	11,671	84.9

stitute (PSI) between March 1984 and December 1998, in a joint effort between the Division of Radiation Medicine at the PSI and the University Eye Clinic (Hôpital Ophthalmique Jules Gonin) in Lausanne. Patients were referred from Western Europe; their countries of residence are listed in Table 1. All were initially evaluated at the University Eye Clinic in Lausanne. None of these patients had received any previous treatment. Work-up included a detailed fundus drawing; standard and wide-angle fundus photography; and A- and B-scan ultrasonography, including measurement of eye length, tumor thickness, and imaging of tumor shape and location. Patients with small posterior tumors underwent fluorescein angiography. Patients' demographic parameters are listed in Table 2, their ophthalmologic parameters in Table 3. Data concerning size and location of tumors, as obtained from ophthalmologic work-up and treatment planning procedure, are listed in Table 4, where tumor size was defined as follows:

S = small: diameter $\leq$ 10 mm and thickness $\leq$ 3 mm

M = mid-sized: 10 mm $<$ diameter $\leq$15 mm and/or 3 mm $<$ thickness $\leq$ 5 mm

L = large: 15 mm $<$ diameter $\leq$ 20 mm and/or 5 mm $<$ thickness $\leq$ 10 mm

XL = extra large: diameter $>$ 20 mm and/or thickness $>$ 10 mm

Table 2. Patients' demographic parameters

		No. of patients	%
Gender	Male	1,188	48.85
	Female	1,244	51.15
Age	≤20 years	25	1.03
	21–30 years	136	5.59
	31–40 years	264	10.85
	41–50 years	485	19.94
	51–60 years	630	25.90
	61–70 years	604	24.84
	71–80 years	248	10.20
	>80 years	40	1.64

Table 3. Ophthalmologic parameters

		No. of eyes	%
Known history of glaucoma	No	2,401	98.6
	Yes	34	1.4
Presence of any kind of hemorraghe	No	2,160	88.7
	Yes	275	11.3
Amount of retinal detachment	Absent	687	28.2
	< 1 Quarter	639	26.2
	1 Quarter	436	17.9
	2 Quarters	593	24.4
	3 Quarters	75	3.1
	Total	5	0.2
Ocular melanocytosis	No	2,350	96.5
	Yes	80	3.3
	No indication	5	0.2
Eye fundus	Visualized	2,404	98.7
	Not visualized	31	1.3

After confirmation of the diagnosis, patients were accepted for PBRT if they met following inclusion criteria: (a) tumor height $<$ 15 mm, and (b) no total retinal detachment.

Exceptions were made to these rules when patients refused enucleation or when the melanoma was located in a patient's only functional eye. Patients accepted for PBRT underwent a surgical procedure for precise demarcation of the tumor margins. The tumor base was localized by transillumination of the globe and indirect ophthalmoscopy. Three to seven tantalum clips (diameter 2.5 mm, thickness 0.5 mm) were sutured onto the outer surface of the sclera to mark the border of the tumor base, as perceived from its shadow during transillumination. Clip surgery was performed by the same surgeon in all cases.

Patients usually left the hospital 2 days later and went to the PSI, where a custom-molded head holder containing a bite block was built for immobilization. After the molding procedure, the patient was installed in a positioning chair developed by the PSI and had a simulation session to obtain data for treatment planning. Sets of orthogonal x-ray pictures showing the tantalum clips in a beam coordinate frame were obtained while the patient fixed his gaze on a small target light set at a known angle. The relative positions of the clips could be accurately measured from these films.

Treatment was planned using the EYEPLAN program developed originally by Goitein and Miller (22), and modified later by Perret (23). In the fall of 1993, we introduced another version of the EYEPLAN program, containing a new eyelid model developed by Sheen (24), and began to use it routinely from 1994. In the original EYEPLAN version of Goitein, the eyelid was modeled as a plane perpendicular to the beam axis intersecting the eyeball. The only adjustable parameter was the intersection point along the beam axis. The EYEPLAN version of Sheen allowed for a more detailed modeling of the eyelid, taking into account

Table 4. Tumor data

		No.	%
Tumor size	S	126	5.2
	M	581	23.9
	L	1,185	48.7
	XL	543	22.3
Extrascleral extension	Present	119	4.9
Location of anterior tumor margin	Iris	97	4.0
	Ciliary body	669	27.5
	Anterior choroid	665	27.3
	Posterior choroid	1,004	41.2
Location of posterior tumor margin relative to optic disc	Infiltration of optic disc	148	6.1
	In contact	306	12.6
	> 0 mm but < 4 mm	728	29.9
	4 mm or more	1,253	51.4
Location of posterior tumor margin relative to macula	Infiltration of macula	378	15.5
	In contact	209	8.6
	> 0 mm but < 4 mm	721	29.6
	4 mm or more	1,127	46.3

differences of lid thickness encountered by the proton beam across the treatment field.

Using clip coordinates as measured on simulation x-ray pictures and ultrasound eyelength, the EYEPLAN program built a two-sphere model of the eye. The shape of the tumor base as defined by the clip position and the fundus photographs was drawn into this eye model, as well as the tumor profile given from the B-scan ultrasound. Consequently, an optimal treatment position was searched for, i.e., a position allowing complete irradiation of the tumor surrounded by a safety margin, while minimizing irradiation of optic disc and nerve, macula, ciliary body, and lens, in descending priority. Maximal penetration depth of proton beam, required width of the spread-out Bragg peak (SOBP), and shape of the aperture corresponding to the tumor shape in treatment position surrounded by a safety margin of usually 2 mm were also determined using the EYEPLAN program. The safety margin was reduced in 34 cases to avoid irradiation of the optic disc and nerve. In cases of unpigmented melanomas, large flat melanomas, and melanomas accompanied by a vitreous hemorrhage obscuring tumor borders, the safety margin was increased (232 cases). After 1993, we also increased the safety margin in the ciliary body part when infiltrated by tumor (102 cases). Beam range was usually selected to place the 90% isodose of the distal falloff 2.5 mm beyond the tumor. The 90% isodose of the proximal SOBP was placed 2.5 mm in front of the tumor. The collimator border defined the location of the 50% isodose. In cases requiring irradiation of the optic nerve, the length of nerve irradiated was minimized by the use of a wedge-shaped absorber placed in the beam proximal to the front surface of the eye, to partially compensate for the spherical surface of the eye. Such a wedge was used in 441 cases.

Table 5. Total physical dose (in Gy) applied in function of treatment period, without consideration of any radiobiological effectiveness

Total physical dose	Year			Total no. of eyes	
	1984–87	1988–93	1994–98		
40–53 Gy	8	14	0	22	0.9%
54.5 Gy	265	1081	986	2332	95.8%
57.3 Gy	28	0	0	28	1.1%
59.1 Gy	10	0	0	10	0.4%
63.6 Gy	13	27	3	43	1.8%
Total				2435	100%

Treatment was delivered in 4 fractions, usually on 4 consecutive days. Most of the patients were treated with a proton dose of 54.5 Gy, which corresponds to 60 CGE (Co-60-Gy equivalent) (14) using a relative biologic effectiveness (RBE) of 1.1. Details of the applied dose are given in Table 5. In 1984, treatment started with the application of 70 CGE in 4 fractions. This dose was lowered to 65 CGE in a first step, then to 63 CGE, and finally to 60 CGE. Some selected patients with a melanoma located close to the optic disc in their only functional eye were treated with a reduced dose, to try to reduce the complication probability.

After treatment, patients were asked to return to the University Eye Clinic of Lausanne for follow-up examinations at intervals of 6 months, 1.5 years, 3 years, 5 years, and then every 2.5 years, or at shorter intervals if required by the individual situation. When patients were unable to return to Lausanne for their follow-up examination (mainly patients living more than 500 km from Lausanne), follow-up data were collected by contacting the referring ophthalmologist.

Follow-up data available through September 1999 were used for this analysis. The number of follow-up examinations and percentage of examinations realized in Lausanne are shown in Table 1, categorized by the different countries of patients' residence. A total of 11,671 follow-up examinations were registered, 85% of them in Lausanne. Follow-up time ranged from 0 to 175 months (mean 48.5 months, median 40 months). Two hundred four patients (8.4%) were lost to follow-up. Patients lost to follow-up were censored at their last available examination.

The endpoint events assessed for each patient included local recurrence and mortality. Local recurrence was defined as a persisting increase of at least 1.0 mm in tumor basal diameter observed by comparison of initial and follow-up fundus photographs and/or in tumor thickness measured by ultrasonography. Increase in tumor thickness was detected in 151 eyes (6.2%). In cases where it was associated with vitreous hemorrhage, tumor bleeding, or exsudative retinal detachment, the patient was observed for some months to confirm suspicion of tumor growth. Ninety-five of 151 patients had such conditions. In 83 of them, the tumor shrunk again after 6 months to 2 years of observation without additional treatment. The 12 remaining patients, having no further tumor growth, but no shrinkage either, are still being observed. Fifty-six of 151 were considered as recurrences. In total, 73 cases of recurrences were diagnosed.

Location and extension of the recurrent site were superimposed onto the initial treatment plan. Dose distribution calculation, dosimetry, and position control X-ray pictures in the treatment record were analyzed to determine if, in retrospect, any of these aspects might have been inadequate.

Local tumor control and survival rates were estimated using the Kaplan–Meier method. Cox proportional hazards modeling was used to assess the prognostic value of various demographic, ophthalmologic, and radiotherapeutic parameters for duration of local control until recurrence and for survival. The Kaplan–Meier method was used to estimate survival in subgroups of patients with and without recurrence, as well as to determine rates of local tumor control and/or survival in different patient subgroups. In these cases, survival and local tumor control rates were compared using the log–rank test. All statistical analysis calculations were done with the SPSS software package version 7.5.

RESULTS

Local tumor control

The actuarial rate of local tumor control for the entire group of patients was 95.8 $\pm$ 0.5% at 5 years and 94.8 $\pm$ 0.7% at 10 years (Kaplan–Meier estimate). Two recurrences occurred after 10 years, 1 at 127 months, and 1 at 130 months, leading to an actuarial local tumor control rate of 92.4 $\pm$ 1.8% at 130 months. No further recurrences were observed up to 175 months of follow-up. Time between PBRT and recurrence diagnosis ranged from 3 to 130 months. Local tumor control failures were diagnosed in 73 eyes. Of these, 51 cases showed tumor regrowth immediately adjacent to the irradiated volume. In 29/51 cases, the growing tumor was localized in the distal part of the target volume, in 22/51 cases, in the ciliary body. Eleven of 73 tumors showed continued growth within the original target volume. Four of 73 recurrences grew totally outside the target volume. For these 4 cases, it is questionable whether they are to be considered as a recurrence or a new tumor. Finally, 7 local tumor control failures were diagnosed and treated outside Lausanne. No further indications could be obtained.

Among these 73 recurrences, 35 were managed by enucleation, 26 received a second PBRT, and 5 were treated with brachytherapy (3 with Ru-106, 1 with Co-60, and 1 with I-125); 2 patients received photocoagulation, and 5 patients received no treatment because of known metastases and poor general health condition. Four patients who received a second conservative treatment showed further tumor growth. One patient who had received photocoagulation was treated again conservatively with Co-60 brachytherapy; 3 patients who had received PBRT twice were enucleated.

The review of all treatment data and their statistical analysis lead to the identification of the following risk factors for local tumor recurrence after PBRT:

Reduced safety margins

Nine recurrences (26.5%) occurred among 34 eyes treated with reduced safety margin, vs. 64 recurrences (2.7%) among 2,401 eyes treated with standard or increased safety margin. The local tumor control rate for these 2 groups at 5 years is 71.3 $\pm$ 8.2% (group with reduced safety margins) and 96.3 $\pm$ 0.5% (group with standard safety margins). The Kaplan–Meier curves are shown in Fig. 1. The Cox proportional hazard model confirmed this influence of reduced safety margins to contribute significantly to reduced local tumor control rate (Table 6). Consequently, in 1988, we abandoned the use of reduced margins.

Large ciliary body tumors

In the entire group of 73 recurrences, 22 appeared in the ciliary body, of which 14 were identified to be ring melanomas. The local tumor control rate in function of the location of the anterior tumor border was further analyzed. Tumors located in the anterior choroid and ciliary body are usually larger than posterior tumors. We analyzed the local control rate for tumors having their anterior margin behind the equator, in the anterior choroid, in the ciliary body, and in the iris, in function of tumor size. For small tumors, there was no difference in local control rates in function of the location of their anterior margin. The same was found for mid-sized tumors. For large and extra-large tumors, however, local control rate was reduced for tumors having their anterior margin in the ciliary body or iris, compared to those with their anterior margin posterior to the ora serrata. Figure 2 shows the local tumor control rate for tumors having their anterior border within the ciliary body in function of the tumor size. Review of the recurrence data suggests that large and extra-large tumors have an increased tendency to build so-called ring melanomas, compared to tumors of smaller size. The Cox proportional hazard model confirmed this influence of tumor size and location to contribute significantly to reduced local tumor control rate (Table 6). For this reason, from 1993 onward, we increased the safety margin of that part of the target volume located in and adjacent to the ciliary body.

Eyelid thickness

Since 1989, we knew that an eyelid within the irradiation field could be the cause of local tumor regrowth in the distal part of the target volume due to an inadequate estimation of eyelid thickness, leading to a reduced range not covering the whole target volume. The simple eyelid model implemented in the EYEPLAN program did not allow for an adequate modeling of the eyelids' varying thickness. This influence was sometimes underestimated and resulted in a range too short to cover the necessary target volume. We discussed this problem with various investigators, especially with M. Sheen from the Clatterbridge Center of Oncology. He developed a new eyelid model, which allowed for a more precise treatment planning. This model was made available to us at the end of 1993, and has been routinely used for all patients treated since 1994.

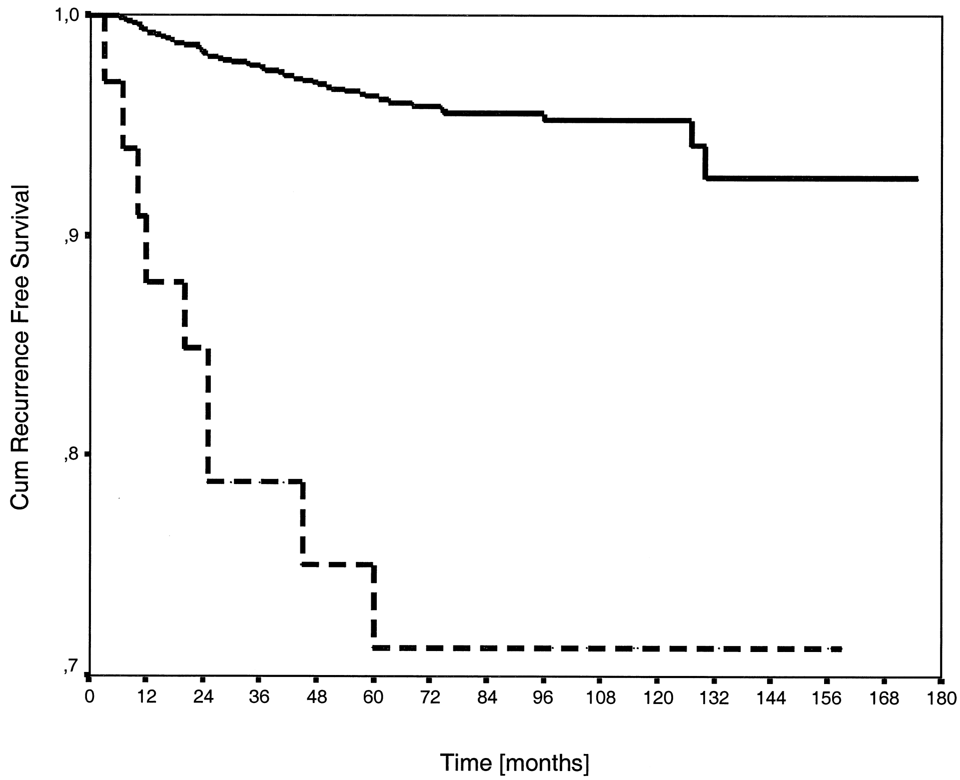

Fig. 1. Local tumor control in function of safety margin. Solid line = standard safety margin, dashed line = reduced safety margin.

Inadequate positioning of tantalum clips

A detailed review of all documents related to the treatment showed that posterior tumors treated before 1988 were generally delimited by only three clips located all on the same side of the tumor, their other side being delimited by the optic disc. Differences between the tumor model from the treatment plan and the fundus photographs were discovered. The distorted vision of patients with macular tumors, making their gaze angle uncertain as well as the location of the clips relative to the optic disc, was identified to be the cause of inappropriate tumor modeling. For this reason, we decided that clips were to be placed all around the tumor if possible, not only on one or two sides, to better localize the target volume; therefore, all sides of the tumor would be delimited by clips. More weight was also given to the fundus photographs for patients treated since 1988.

Table 6. Local tumor control and survival: Cox proportional hazard model

Variable	Local control *p*-value	Survival *p*-value
Age	0.6244	**0.0000***
Gender	0.0265*	0.4246
Amount of retinal detachment	0.3854	**0.0015***
Ocular melanocytosis	0.3063	**0.0305***
Invisible eye fundus	0.8485	0.2864
Largest tumor diameter	0.5337	**0.0000***
Tumor thickness	0.3228	**0.0044***
Tumor size	**0.0045**	0.2737
Anterior tumor margin in iris	0.3254	0.1684
Anterior tumor margin in ciliary body	0.0021*	0.0734
Anterior tumor margin anterior to equator	0.8090	0.3177
Anterior tumor margin posterior to equator	0.5961	0.8239
Infiltration of optic disc	0.3823	0.6852
Distance of optic disc	0.4448	0.8122
Extrascleral extension	0.9305	**0.0000***
Total dose	0.2059	0.4892
Use of wedges	0.3024	0.4345
Increased safety margin	0.2849	0.2139
Reduced safety margin	**0.0000***	0.8928

* Statistically significant parameters.

Gender

The Cox proportional hazard model identified gender to be of significance. As shown in Fig. 3, women have a better local tumor control rate than men. The log–rank test indicates a *p*-value of 0.0318. In our series, 43 of 1,188 men showed a recurrence vs. 30 of 1,247 women.

The measures introduced in 1988 (no reduced safety margin, circumscribed delimitation of the tumor by clips) and 1993 (increase of safety margin for ciliary body tumors, new eyelid model) correlate with an increase of local tumor control rate, as shown in Fig. 4. The number of patients treated in those time periods and the number of recurrences diagnosed, are listed in Table 7. The presumed causes of the 62 recurrences adjacent to and within the target volume,

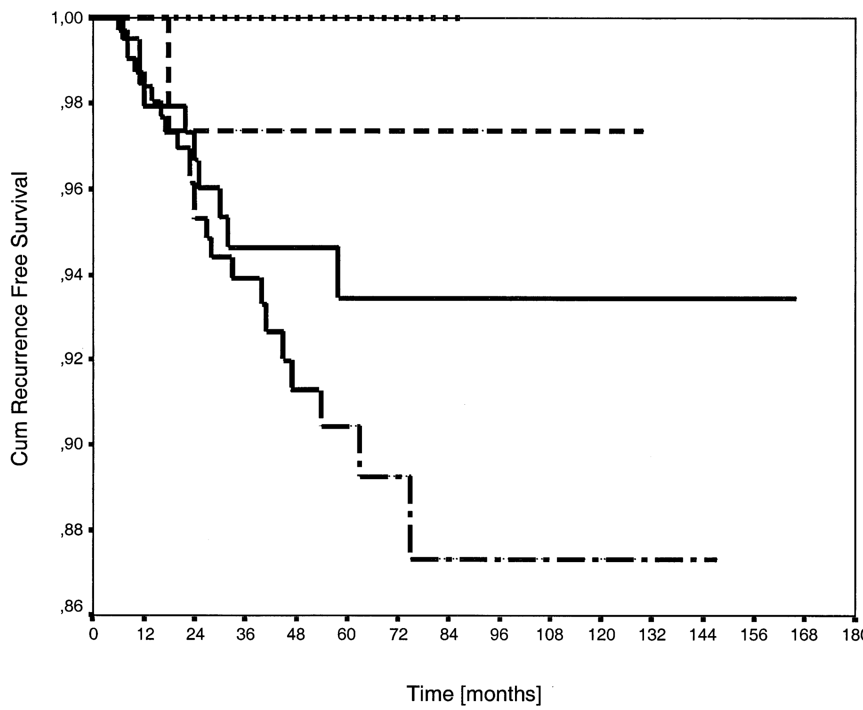

Fig. 2. Local tumor control in function of tumor size for tumors having their anterior margin within the ciliary body. Dotted line = small tumors (S), dashed line = mid-sized tumors (M), Solid line = large tumors (L), and dashed and dotted line = very large tumors (XL).

where all data were available, are listed in Table 8. Because more than one cause may have been at the origin of a local tumor control failure, their total differs from 62. In the 4 cases of distant recurrence, we could not identify any probable cause for the local control failure.

Reduction of tumor size

We investigated the reduction of tumor thickness in function of time. The mean persisting tumor thickness was 81% at 6 months, 65% at 1.5 years, 55% at 3 years, and 48% at 5 years. We also calculated the mean persisting tumor thickness for the

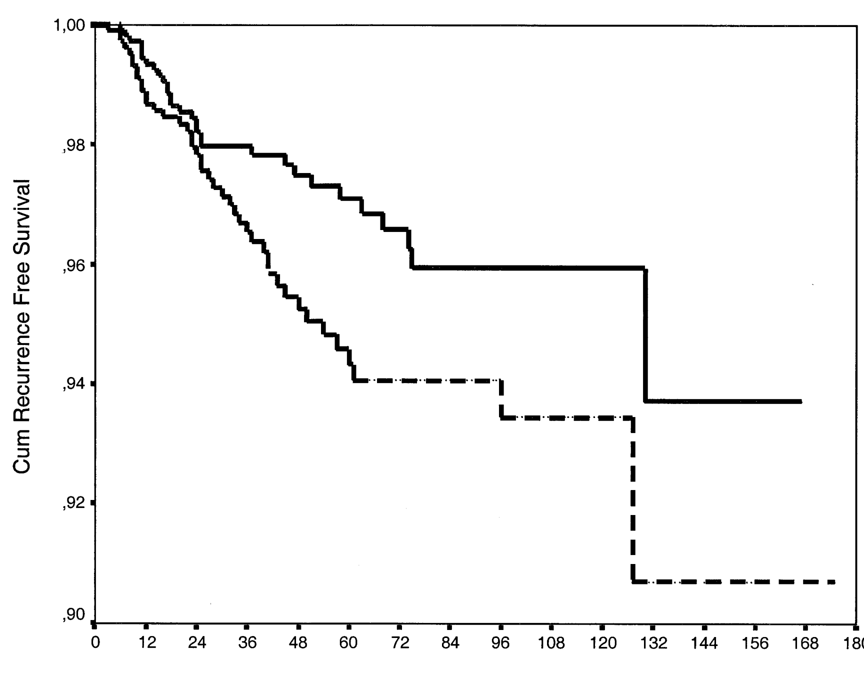

Fig. 3. Local tumor control in function of gender. Solid line = women, dashed line = men.

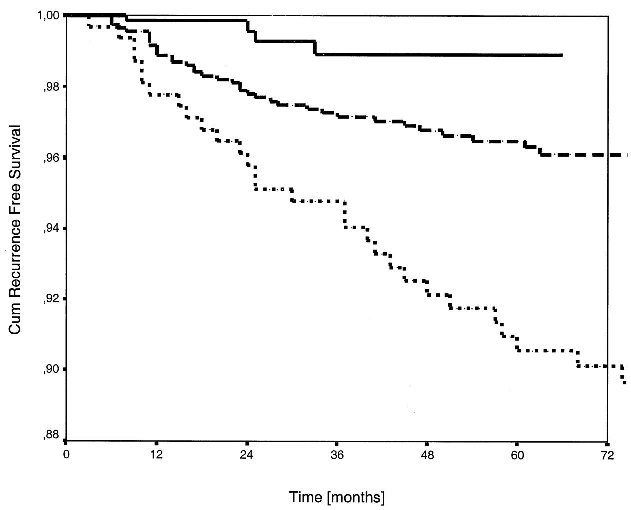

Fig. 4. Local tumor control in function of our experience. Patients treated between March 1984 and December 1987 = dotted line, patients treated between January 1988 and December 1993 = dashed line, and patients treated between January 1994 and December 1998 = solid line.

three groups of patients treated with a dose smaller than 54.5 Gy protons, with 54.5 Gy and with a larger dose (see Table 5) to determine whether a dose dependence of size reduction could be found. The results are shown in Table 9. These data suggest a faster size reduction with higher doses.

Survival

The survival rates were calculated for patients having their tumor controlled after the first treatment and those with recurring tumors. Two hundred seventy-six patients had tumor-related deaths in the group of 2,362 patients with no evidence of local tumor control failure, while 34 tumor-related deaths were registered in the group of 73 patients with recurrent tumors. The log–rank test indicated a p-value of $p = 0.0000$. The Kaplan–Meier survival rates at 10 years were 72.6 ± 1.9% and 47.5 ± 6.5%, respectively. The corresponding survival curves are shown in Fig. 5.

The 5-year survival rates were also calculated separately for the group of patients treated between March 1984 and December 1987, January 1988 and December 1993, and January 1994 and December 1998. They were 81.8 ± 2.2% (97 tumor-related deaths among 323 patients treated), 83.3 ± 1.2% (191 tumor-related deaths among 1,122 patients treated), and 87.9 ± 3.5% (22 tumor related deaths among 990 patients treated), respectively. The corresponding survival curves are shown in Fig. 6. According to the log–rank test, there is no statistically significant difference between the first and second group, while the survival of the patients treated after 1993 is better than the survival of patients treated earlier ($p = 0.0042$ for Group 1 vs. Group 3, and $p = 0.0033$ for Group 2 vs. Group 3).

To investigate the question of whether this improved survival could be attributed to the modifications implemented into our treatment technique or simply to a more favorable selection of patients, we compared these three groups for different parameters influencing the survival rates (see Table 6). The Cox model identified the following parameters to reduce survival probability: advanced patient age, greater amount of retinal detachment, presence of ocular melanocytosis, larger tumor diameter and/or thickness, and presence of an extrascleral extension. Patients' age constantly increased, the mean age of the three groups being 52.3, 53.9, and 55.3 years, respectively. The amount of retinal detachment decreased with time. Tumor diameter was larger in Groups 2 and 3, while tumor thickness progressively decreased. The presence of ocular melanocytosis did not vary from one time period to another. The number of patients with extrascleral extension increased constantly. In conclusion, age, largest tumor diameter, and extrascleral extension—which are the more significant factors—worsened over time, while retinal detachment and tumor thickness—which influence survival less—became more favorable. The mean values of these factors, categorized within time periods, are listed in Table 10.

DISCUSSION

This study presents a series of uveal melanoma patients treated with PBRT in Western Europe, containing the largest number of patients and longest follow-up time.

We compared our results (a final local tumor control rate of 98.9% at 5 years after improvement of the technique) to

Table 8. Identified causes of local tumor control failure

Probable cause	No. of cases
Reduced safety margin	9
Infiltration of ciliary body/ring melanoma	22
Eyelid in irradiation field	20
Clip position/eye model	17
Wrong ultrasound measurement of eyelength	3

Table 7. Number of patients and recurrences

Time	Patients treated	No. of recurrences	%
March 1984 to December 1987	323	32	9.9
January 1988 to December 1993	1122	37	3.3
January 1994 to December 1998	990	4	0.4

Table 9. Tumor thickness reduction in function of applied dose

	< 54.5 Gy (%)	54.5 Gy (%)	> 54.5 Gy (%)	p-value
6 months	86.1	80.9	77.1	0.048
1.5 years	75.8	65.1	59.7	0.023
3 years	56.7	55.7	44.7	0.003
5 years	70.0	48.7	40.8	0.020

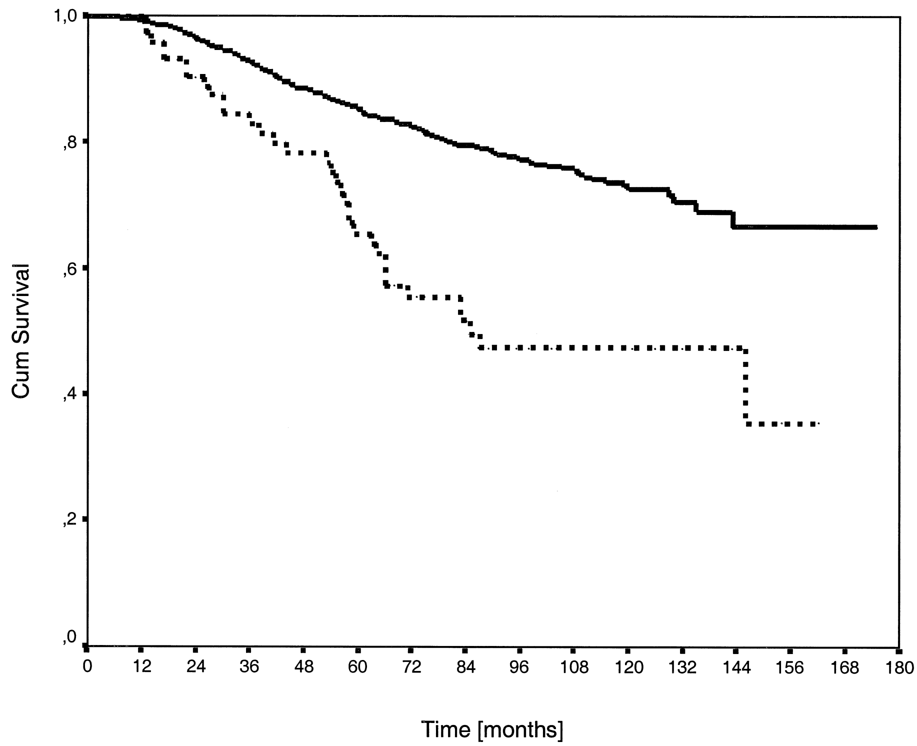

Fig. 5. Survival in function of local tumor control. Solid line = tumors controlled locally, dashed line = recurrent tumors.

those published by other groups treating uveal melanoma with proton or helium beam radiotherapy. The beam and safety margin parameters can be summarized as follows: the distance between the 90% and the 10% isodose is 1.8 mm at the distal and the lateral edge. The distal 90% isodose is placed at 2.5 mm behind the gross tumor volume, while the lateral 90% isodose is placed at 1.1 mm.

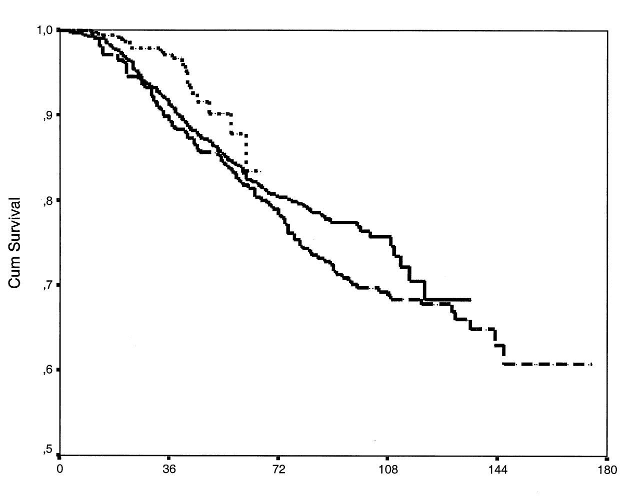

Fig. 6. Survival in function of our experience. Patients treated between March 1984 and December 1987 = dashed line, patients treated between January 1988 and December 1993 = solid line, patients treated between January 1994 and December 1998 = dotted line.

Gragoudas and coworkers (25) reached a local tumor control rate of 96.3% at 5 years. They used a 1.5-mm safety margin around the tumor (gross tumor volume [GTV]) defined by the 90% isodose. Because their proton beam is degraded from 160 MeV, the distance between the 90% and the 10% isodose is 3 mm. Char and coworkers showed a 5-year local tumor control rate of 94.7% (26, 27), using a helium ion beam. Their safety margin was defined by the 90% isodose placed at 2 mm on the lateral sides of the GTV and at 3 mm on the distal part. The distance between the 90% and 10% isodose was 1.3 mm laterally and 7 mm distally (9). Courdi and coworkers found a local control rate of 89% at 6.5 years (28). The dose fall-off at the distal edge from 90% to 10% was within 0.8 mm (29). The collimator border defining the lateral margin was placed at 2.5 mm, which corresponds to a margin of approximately 2 mm, as defined by the 90% isodose. They routinely used a distal safety margin of 2.5 mm defined by the 90% isodose, which was reduced to preserve the optic nerve and/or the macula in a nonspecified number of patients. The sharper fall-off at the distal edge and the use of a reduced distal safety margin might be the cause of the somewhat reduced local tumor control rate obtained by this group. The sharp dose fall-off at the edges of the target volume is, of course, an advantage for minimizing the irradiation of healthy tissue; however, it might be a disadvantage to ensure local tumor control if the safety margin is chosen too tight to the clinical target volume.

These data show that a local tumor control rate of 96%

Table 10. Mean values of parameters influencing survival in function of time

Parameter		Year		
		1984–1988	1989–1993	1994–1999
Age		52.2 years	53.9 years	55.3 years
Amount of retinal detachment	Absent	15.2%	25.5%	35.6%
	<1 Quarter	26.0%	26.8%	25.7%
	1 Quarter	14.9%	15.7%	21.4%
	2 Quarters	32.5%	29.2%	16.2%
	3 Quarters	10.8%	2.5%	1.2%
	Total	0.6%	0.3%	0%
Largest tumor diameter		14.71 mm	16.78 mm	15.89 mm
Tumor thickness		6.54 mm	6.13 mm	6.05 mm
Extrascleral extension		2.8%	5.1%	5.4%
Ocular melanocytosis		4.0%	2.9%	3.5%

and more can be achieved with PBRT and similar techniques at 5 years. This rate will be reduced when using smaller safety margins, as is demonstrated by our results and those of Courdi and coworkers. Our results also suggest that the local control rate can be improved to 99% when all identified risk factors have been eliminated.

Following episcleral plaque brachytherapy, 5-year local tumor control rates between 83 and 92% have been published (6, 16, 19, 30, 31). Tumor size and location significantly influence the rate of local tumor control; large tumors and tumors located close to the optic disc have poorer local control rate. The advantages of PBRT are that local tumor control rate is higher and, in the case of posterior tumors, that it is not influenced by tumor size.

In a recently published study, Gragoudas *et al*. investigated the influence of different applied doses on local tumor control, visual acuity, complications, and survival (32). They found similar results for tumors treated with 50 CGE (Cobalt Gray Equivalent) and 70 CGE applied in five fractions. Their study was limited to relatively small tumors (largest diameter $<$ 15 mm and thickness $<$ 5 mm). Our local tumor control rate of nearly 99% suggests that the dose of 60 CGE applied in four fractions is high enough to ensure local tumor control.

Gragoudas also investigated the regression in tumor height. His data suggest, as do ours, that a faster regression occurs when a higher dose is applied. While this difference is statistically significant in our series of patients, it was not in the study of Gragoudas. However, the series of Gragoudas is limited to small tumors, while our series contains mainly large tumors.

When we introduced PBRT in 1984, this treatment was considered to be a possible alternative to enucleation for tumors located close to the optic disc and macula, and for large tumors. Different articles had suggested that survival was influenced by local tumor control rate (33–35). Karlsson and coworkers (31) investigated the survival rate of uveal melanoma patients treated with Co-60 brachytherapy for controlled and recurrent tumors. They found an increased mortality rate in the group of recurrent tumors, where twice the number of patients died from metastases as compared to the group of controlled tumors; the survival rate at 5 years was 58% for patients with recurrent tumors vs. 82% for patients with controlled tumors. Shields and coworkers investigated the influence on survival in 35 patients with juxtapapillary uveal melanomas treated with brachytherapy (19). They also found an increased mortality among patients with recurrences (42% of tumor-related deaths), compared to patients having their tumor controlled (11% of tumor-related deaths). Our results support the assumption that local tumor control failure is associated with a higher risk for the development of metastases. In our series, while local control at 5 years increased from 90.6% to 96.3% and 98.9% for the three analyzed time periods, the cause-specific 5-year survival rate changed from 81.9% to 83.3% and 87.9%, respectively. An important question is whether the increased survival rate can be attributed to the reduction of local tumor control failures or whether selection criteria are responsible for better survival. As demonstrated previously, three of six prognostic factors became less favorable with time, while two became more favorable and one remained constant. Due to the fact that the three risk factors that worsened are the most significant factors according to the Cox model, one would expect more deaths from metastases among the patients treated after 1988. Because this is not the case, we conclude that the obtained improvement of the survival rate can be attributed to the reduction of local tumor control failures.

CONCLUSION

Our results suggest that PBRT seems to be an excellent treatment for achievement of local tumor control. Our local tumor control rate at 5 years is 98.9% for patients treated since 1994. Risk factors for local tumor control failure were identified as a reduction of the safety margin, large tumors infiltrating the ciliary body, the presence of an eyelid within the irradiation field, inadequate delimitation of the tumor border by tantalum clips, and male gender. Our measures to maximize local control rates were: using standard safety margins in all cases, increasing the safety margin for large ciliary body tumors, introducing a new eyelid model in the treatment planning software, and delimiting all sides of the tumor by clips. We also found a statistically significant correlation between tumor-related mortality and local tumor control failure. The improvement of local tumor control allowed for a better survival rate.

REFERENCES

1. Brady LW, Markoe AM, Amendola BE, *et al*. The treatment of primary intraocular malignancy. *Int J Radiat Oncol Biol Phys* 1988;15:1355–1361.
2. Damato BE, Paul J, Foulds WS. Risk factors for residual and recurrent uveal melanoma after trans-scleral local resection. *Br J Ophthalmol* 1996;80:102–108.
3. Brady LW, Hernandez JC. Brachytherapy of choroidal hemangiomas. *Strahlenther Onkol* 1992;168:61–65.
4. Petrovich Z, Luxton G, Langholz B, *et al*. Episcleral plaque radiotherapy in the treatment of uveal melanomas. *Int J Radiat Oncol Biol Phys* 1992;24:247–251.
5. Lommatzsch PK. Results after β-irradiation (^{106}Ru/^{106}Rh) of choroidal melanomas: 20 years' experience. *Br J Ophthalmol* 1986;70:844–851.
6. Gündüz K, Shields CL, Shields JA, *et al*. Plaque radiotherapy of uveal melanoma with predominant ciliary body involvement. *Arch Ophthalmol* 1999;117:170–177.
7. Tokuuye K, Akine Y, Sumi M, *et al*. Fractionated stereotactic radiotherapy for choroidal melanoma. *Radiother Oncol* 1997; 43:87–91.
8. Zehetmeyer M, Menapace R, Kitz K, *et al*. Stereotactic irradiation of uveal melanoma with the Leksell gamma unit. In: Wiegel T, Bornfeld N, Foerster MH, *et al*., editors. Radiotherapy of ocular disease. Front Radiat Ther Oncol. Basel: Karger; 1997;30:47–55.
9. Saunders WM, Char DH, Quivey JM, *et al*. Precision, high dose radiotherapy: Helium ion treatment of uveal melanoma. *Int J Radiat Oncol Biol Phys* 1985;11:227–233.
10. Linstadt D, Castro J, Char D, *et al*. Long-term results of helium ion irradiation of uveal melanoma. *Int J Radiat Oncol Biol Phys* 1990;19:613–618.
11. Gragoudas ES, Seddon JM, Egan K, *et al*. Long-term results of proton beam irradiated uveal melanomas. *Arch Ophthalmol* 1987;94:349–353.
12. Munzenrider JE. Proton therapy for uveal melanomas and other eye lesions. *Strahlenther Onkol* 1999;175(Suppl. II):68–73.
13. Augsburger JJ, Correa ZM, Freire J, *et al*. Long-term survival in choroidal and ciliary body melanoma after enucleation versus plaque radiation therapy. *Ophthalmology* 1998;105: 1670–1678.
14. Gragoudas ES, Goitein M, Verhey L, *et al*. Proton beam irradiation: An alternative to enucleation for intraocular melanomas. *Ophthalmology* 1980;87:571–581.
15. Seddon JM, Gragoudas ES, Albert DM, *et al*. Comparison of survival rates for patients with uveal melanoma after treatment with proton beam irradiation or enucleation. *Am J Ophthalmol* 1985;99:282–290.
16. Char DH, Quivey JM, Castro JR, *et al*. Helium ion versus iodine 125 brachytherapy in the management of uveal melanoma. *Ophthalmology* 1993;100:1547–1554.
17. Char DH, Kroll S, Quivey J, *et al*. Long term visual outcome of radiated uveal melanomas in eyes eligible for randomisation to enucleation versus brachytherapy. *Br J Ophthalmol* 1996;80:117–124.
18. Wilson MW, Hungerford JL. Comparison of episcleral plaque and proton beam radiation therapy for the treatment of choroidal melanoma. *Ophthalmology* 1999;106:1579–1587.
19. De Potter P, Shields CL, Shields JA, *et al*. Impact of enucleation versus plaque radiotherapy in the management of juxtapapillary choroidal melanoma on patient survival. *Br J Ophthalmol* 1994;78:109–114.
20. Straatsma BR, Fine SL, Earle JD, *et al*. Enucleation versus plaque irradiation for choroidal melanoma. *Ophthalmology* 1988;95:1000–1004.
21. Jensen OA. Malignant melanomas of the human uvea: 25 year follow-up of cases in Denmark, 1943–1952. *Acta Ophthalmol* 1982:60:161–182.
22. Goitein M, Miller T. Planning proton therapy of the eye. *Med Phys* 1983;10:275–283.
23. Perret C. The therapy planning program EYE. Swiss Institute for Nuclear Research, Internal Report; 1987.
24. Sheen M. EYEPLAN proton therapy planning program. Clatterbridge Centre for Oncology, Internal Report; 1995.
25. Munzenrider JE, Verhey LJ, Gragoudas ES, *et al*. Conservative treatment of uveal melanoma: Local recurrence after proton beam therapy. *Int J Radiat Oncol Biol Phys* 1989;17: 493–498.
26. Char DH, Kroll SM, Castro J. Long-term follow-up after uveal melanoma charged particle therapy. *Trans Am Ophthalmol Soc* 1997;95:171–187.
27. Char DH, Kroll SM, Castro J. Ten-year follow-up of helium ion therapy for uveal melanoma. *Am J Ophthalmol* 1998;125: 81–89.
28. Courdi A, Caujolle JP, Grange JD, *et al*. Results of proton therapy of uveal melanomas treated in Nice. *Int J Radiat Oncol Biol Phys* 1999;45:5–11.
29. Chauvel P, Sauerwein W, Bornfeld N, *et al*. Clinical and technical requirements for proton treatment planning of ocular disease. In: Wiegel T, Bornfeld N, Foerster MH, *et al*., editors. Radiotherapy of ocular disease. Front Radiat Ther Oncol. Basel: Karger; 1997;30:133–142.
30. Gündüz K, Shields CL, Shields JA, *et al*.Radiation complications and tumor control after plaque radiotherapy of choroidal melanoma with macular involvement. *Am J Ophthalmol* 1999; 127:579–589.
31. Karlsson UL, Augsburger JJ, Shields JA, *et al*. Recurrence of posterior uveal melanoma after ^{60}Co episcleral plaque therapy. *Ophthalmology* 1989;96:382–388.
32. Gragoudas ES, Lane AM, Regan S, *et al*. A randomized controlled trial of varying radiation doses in the treatment of choroidal melanoma. *Arch Ophthalmol* 2000;118:773–778.
33. Suit HD. Potential for improving survival rates for the cancer patient by increasing the efficacy of treatment of the primary lesion. *Cancer* 1982;50:1227–1234.
34. Suit HD, Westgate SJ. Impact of improved local control on survival. *Int J Radiat Oncol Biol Phys* 1986;15:453–458.
35. Suit HD. Local control and patient survival. *Int J Radiat Oncol Biol Phys* 1992;23:653–660.